

Liam Barkley

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Academics

- Orono High School, Class of 2027
- GPA: 4.536
- ACT: 35
- AP Scholar with Distinction
- University of Minnesota PSEO student studying advanced topics in mathematics
- CPR certified, November 2023

Athletics

- Orono Nordic Skiing: Varsity skier, grades 10–11; summer member of Minneapolis Ski Club
- Orono Cross Country: Varsity runner, grades 10–12; state qualifier and third-place team finisher junior year
- Orono Track and Field: Varsity runner, grades 9–12; all-state runner; state qualifier and 8th-place medalist with the 4x800 relay team

Leadership and Activities

- Orono Robotics Team: Design, programming, and drive teams; Programming Team Lead for both years on the team
- Orono Student Senate: Sophomore and Junior Senator
- Orono Newspaper: Features Editor and Staff Writer
- Orono Math Team: First-year letterer
- Orono Wind Ensemble: Band leadership member
- Metro-West All-Conference Honor Band: Oboist

Volunteering and Service

- Community STEM Outreach: Help the robotics team connect with the community through robotics events
- Feed My Starving Children: Participated in multiple service events with a church group
- National Honor Society and Tri-M Music Honor Society

Skills

- Programming: C++ and Python experience; currently learning MATLAB and Simulink
- Design: Fusion 360 - Onshape - EasyStone; CAD, CAM, PCB design, and simulation experience
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Personal STEM Projects and Research

- Developed custom flight software, vehicle designs, and simulations for multiple thrust-vector-controlled scale rockets
- Mach cutoff research: Studied supersonic shockwave refraction during slow supersonic flight and explored its potential use in a real-time guidance system
- Liquid rocket engine: Studied supersonic shockwave refraction during slow supersonic flight and explored its potential use in a real-time guidance system

Work Experience

- Dakota Granite: Machined various support hardware using CNC mills and lathes, and helped bring machines online during summer 2025 (mid-June–early August) and late December 2025
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